



Application of full-factorial design in the synthesis of polypropylene-g-poly(glycidyl methacrylate) functional material for metal ion adsorption

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ABSTRACT

The graft polymerization of glycidyl methacrylate onto electron beam pre-irradiated polypropylene (PP) nonwoven fabric was optimized using a 4^3 full factorial design analysis. The analysis yielded a polynomial equation that relates the linear, quadratic and interaction effects of the independent parameters to degree of grafting (Dg). The linear terms (i.e. absorbed dose, reaction time and monomer concentration), quadratic terms of time and concentration, and interaction term between absorbed dose and time were determined as significant independent parameters based from analysis of variance (ANOVA). The optimum grafting time and absorbed dose to achieve 150% Dg at 5% monomer concentration were 3.5 h and 39.8 kGy, respectively. The pristine PP, polypropylene-g-poly(glycidyl methacrylate) (PP-g-PGMA) and functionalized grafted materials were characterized using ATR-FTIR spectroscopy, thermogravimetric analysis and scanning electron microscopy. The affinities of the synthesized adsorbents towards the target metal ions at pH 4 were established to be in the following order: Cr(VI) > Pb(II) ~ Cd(II) for the amine functionalized PP-g-PGMA; and Pb(II) > Cd(II) > Cr(VI) for the carboxylic acid functionalized PP-g-PGMA.

1. Introduction

In many countries in the Asia and Pacific Region, including the Philippines, industrial wastes in the form of toxic metals, organic compounds, dyes and microorganisms are significant sources of water pollution. These wastes have detrimental health and environmental effects, therefore treatment of waste and process waters containing significant amount of these materials is of utmost importance. The general separation of heavy metals from aqueous wastes is normally carried out using chemical precipitation techniques, usually transforming them into their hydroxide form. However, hydroxide precipitation alone is not sufficient to decrease the heavy metal concentration and meet the limit indicated in the regulation. Moreover, the large amount of waste sludge formed from the precipitation process is also another source of problem in the treatment process. In this regard, membrane and ion exchange techniques are normally used to bring down the concentration to the allowable limits. Previous studies showed that radiation grafted polymers were capable of reducing heavy metal ion concentrations to ppb level (Sokker et al., 2009; Wojnarovits et al., 2010; Madrid et al., 2013); sufficient to meet the criteria set by the Philippines' environmental agency regulation.

Radiation-induced graft polymerization can impart desired functional groups on otherwise unreactive trunk polymers (e.g. polyethy-

lene, PE, and polypropylene, PP) without the use of chemical initiators. Initiation of graft polymerization using electron beam or gamma irradiation is more desirable than chemical initiation because it does not leave chemical initiator residues in the synthesized material and the polymerization can be performed at ambient conditions. The process is generally simple, easy to control and can be applied to trunk polymer materials of various sizes and forms (e.g. powder, fibers, films, etc). The use of PE or PP nonwoven fabrics allows the application of higher feed flow rate which increases the amount of waste water that can be processed per unit time, leading to lower treatment cost and higher productivity (Jyo et al., 2001; Sekine et al., 2010). These synthetic NWFs have desirable features that make them suitable as trunk materials in synthesis of adsorbents for waste water treatment. These features include chemical inertness, high mechanical strength, low resistance to liquid flow and high surface area leading to high adsorbent loading (Nasef and Guven, 2012). Also, carrying out grafting with the monomer in emulsion state is an environmental merit because the grafting mixture contains more than 90% water, hence avoiding the use of too much organic solvent that are normally utilized in typical grafting processes.

The application of design of experiments provides a wider analysis of how the different factors affect a target parameter, instead of the traditional one-variable-at-a-time experiments, which were used by

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majority of researchers. Experimental design techniques, such as full factorial design, central composite design and Box-Behnken design are useful tools in the characterization of various industrial and analytical processes by studying the effects and interactions of variables affecting them. Nowadays, factorial designs have proved their applicability and are widely utilized in the statistical planning of experiments to obtain empirical models relating process factors to process response. The factorial design method determines which independent parameters have significant effect on the response variable as well as how the effect of one independent parameter varies according to the level of other factors (Passos et al., 2006). In the field of radiation-induced graft polymerization, only few studies that utilized design of experiments were reported (Arayaprane and Rempel, 2004; Nasef et al., 2012).

This paper reports the optimization of reaction parameters on the electron beam-induced grafting of glycidyl methacrylate onto PP using a 4^3 full factorial design analysis. The PP-g-PGMA was functionalized to prepare amine and carboxylic type adsorbents, and the Cr(VI), Pb(II) and Cd(II) adsorption capabilities of the functionalized materials were measured.

2. Materials and methods

2.1. Materials

The polypropylene nonwoven fabric (PP NWF) that was provided by Agriculture & Supplies Corp. (Philippines) was used as trunk polymer. Glycidyl methacrylate monomer (GMA, $\geq 99.7\%$, Aldrich), polyoxyethylene sorbitan monolaurate (Tween 20), ethylenediamine (EDA, $\geq 99\%$, Sigma-Aldrich), iminodiacetic acid (IDA, $> 98\%$, Aldrich), potassium dichromate ($K_2Cr_2O_7$, $\geq 99.9\%$, Vetec), lead acetate trihydrate ($Pb(CH_3COO)_2 \cdot 3H_2O$, $> 99.5\%$, Techno Pharmachem), cadmium acetate dihydrate ($Cd(CH_3COO)_2 \cdot 2H_2O$, $> 98\%$, Sigma-Aldrich), dimethylsulfoxide (DMSO, RCI Labscan), isopropanol (IPA, $> 99.8\%$, RCI Labscan) and methanol ($> 99.9\%$, RCI Labscan) were all used as received.

2.2. Irradiation of PP NWF

PP samples (2 cm $\times$ 2 cm) were placed in polyethylene resealable bags. Prior to irradiation, the air inside the polyethylene bags was displaced with nitrogen gas. The samples were irradiated to the desired dose using electron beam (2.5 MeV energy and 6.5 mA current) at room temperature.

2.3. Grafting of GMA onto PP and functionalization reactions

Factorial design allows the simultaneous study of the effects that several parameters may have on an optimization of a particular process. In a full-factorial design, responses are measured at all combinations of the experimental parameter levels. A four-level full-factorial model was utilized in this study. The degree of grafting (Dg) was chosen as the response parameter and the absorbed dose, reaction time and monomer concentration were selected as the factors/independent parameters. Each factor was studied on four levels (i.e. 4^3 full factorial design) and the values of the levels for each factor were pre-determined based from previous experiments (Table 1).

The GMA monomer was mixed with Tween 20 and water to prepare

Table 1

Factors and levels for the optimization experiment using 5:1 monomer-to-surfactant weight ratio.

Experimental Factors	Level 1	Level 2	Level 3	Level 4
Absorbed dose, kGy	10	20	30	40
Reaction time, h	1	2	3	4
GMA Concentration, %(wt/wt)	3	5	7	9

emulsions containing different monomer concentrations (e.g. 3%, 5%, 7% and 9% (wt/wt) GMA) at 5:1 monomer to surfactant weight ratio. The monomer mixture was homogenized at a rate of 4800 rpm for 15 min and then deoxygenated by nitrogen bubbling for 1 h. The irradiated PP was transferred in a glass ampoule and the air inside the ampoule was immediately evacuated using a vacuum line. Afterwards, the deoxygenated GMA emulsion was drawn into the glass ampoule. The grafting reaction was carried out in a thermostatic water bath at 40 °C for 1–4 h. The polypropylene-g-poly(glycidyl methacrylate) (PP-g-PGMA) samples were repeatedly washed with methanol to remove non-reacted monomers, then dried *in vacuo* overnight at 40 °C.

The degree of grafting (Dg) was determined gravimetrically using Eq. (1):

$$Dg = \frac{W_g - W_i}{W_i} \times 100 \quad (1)$$

where w_i and w_g are the weights of the PP before and after grafting, respectively. Five samples were grafted for each combination of grafting parameters and the average Dg was reported.

Amine functional groups were introduced to the PP-g-PGMA by ring-opening reaction of the epoxy groups with EDA. PP-g-PGMA samples with approximately 150% Dg were added to a glass ampoule containing 100 g solution of EDA in isopropanol. Amination of grafted samples was carried out for 15–180 min in a thermostatic water bath at 60 °C. The amine functionalized PP-g-PGMA was repeatedly washed with methanol and then dried in a vacuum oven. The amine functional group density was calculated using Eq. (2):

$$\text{amine group density (mmol/gram - adsorbent)} = \left(\frac{w_f - w_g}{w_f} \right) \times \frac{1000}{MW} \quad (2)$$

where w_g and w_f are the weights of PP-g-PGMA before and after functionalization, respectively, and MW is the molar mass of EDA. Three samples were functionalized and the average amine group density was reported.

Besides amine group, carboxylic acid is another functional group capable of adsorbing metal ions via chelation or ion exchange. PP-g-PGMA samples with approximately 150% Dg were reacted with 100 mL of 0.4 M IDA solution to impart carboxylic functional groups on the olefinic trunk material. The reaction was carried out in a thermostatic water bath at 80 °C for 24 h. The carboxylated PP-g-PGMA was stirred in water for 4 h at room temperature, and then washed repeatedly with methanol. After drying *in vacuo*, the IDA functional group density was calculated using Eq. (2). Three grafted samples were functionalized and the average IDA group density was reported.

2.4. Response fitting

The 4^3 full factorial design required 64 experiments. The average of five runs of the independent parameters in correlation with the responses was recorded. The experimental design and analysis of data were performed using commercial statistical software Design Expert Version 9 (Stat-Ease, Inc., Minneapolis, MN). The software was used to fit the responses to a quadratic polynomial regression equation that relates the linear, quadratic and interaction effects of the independent parameters to Dg. The software was also used to produce 3D-surface and 2D-contour plots and to perform ANOVA and optimization using the desirability function.

2.5. Batch adsorption

Metal ion solutions (5–1000 ppm initial concentration) were prepared by dissolving the metal salts in deionized water. A weighed piece of the functionalized PP-g-PGMA was added to each 50-mL metal ion solution. The batch adsorption studies were conducted in a continuously stirring batch process at room temperature. After 24 h, the

adsorbent was removed and the solution was filtered through a 0.45 μm syringe filter. The metal ion concentration was measured before and after adsorption using inductively-coupled plasma atomic emission spectrometry (ICP-AES) (Shimadzu ICPS 7510). The amount of adsorbed metal ion was calculated according to Eq. (3):

$$q_e = \frac{(C_o - C_e)V}{M} \quad (3)$$

where q_e (mg/g) is the amount of metal ion adsorbed, C_o and C_e are the initial and final concentration (ppm) of the metal ion in aqueous phase, respectively, V is the volume of solution (L) and M (g) is the mass of functionalized PP-g-PGMA.

2.6. Effect of initial pH

The effect of metal ion solution's initial pH on the removal of Cr(VI), Cd(II) and Pb(II) was evaluated. Fifty mL metal ion solutions, each with 50 ppm initial concentration, were prepared by dissolving the metal salts in deionized water. The pH of the solution was adjusted with 0.1 M HNO_3 or 0.1 M NaOH and the pH value was monitored using a glass electrode Oakton pH meter. A weighed piece of the functionalized PP-g-PGMA was added to each metal ion solution. The adsorption process was carried out with continuous stirring at room temperature. After 24 h, the adsorbent was removed and the solution was filtered through a 0.45 μm syringe filter. The supernatant solutions, before and after adsorption, were analyzed with ICP-AES for metal ion content. The amount of adsorbed metal ion was calculated using Eq. (3).

2.7. Competitive adsorption

Fifty mL binary metal solutions comprised of (a) 50 ppm each of Cr(VI) and Cd(II) and (b) 50 ppm each of Cd(II) and Pb(II) were prepared. The initial pH of both solutions was adjusted to pH 4. A weighed piece of functionalized PP-g-PGMA was added into the binary metal ion solution. The adsorption was carried out with continuous stirring at room temperature. After 24 h, the adsorbent was removed and the solution was filtered through a 0.45 μm syringe filter. The supernatant solutions, before and after adsorption, were analyzed with ICP-AES for metal ion content. The amount of adsorbed metal ion was calculated using Eq. (3). An attempt to study a tertiary metal system for competitive adsorption was prevented by the reaction between $\text{K}_2\text{Cr}_2\text{O}_7$ and $\text{Pb}(\text{CH}_3\text{COO})_2 \cdot 3\text{H}_2\text{O}$, which lead to precipitate formation.

2.8. Characterization techniques

The infrared spectra of pristine, grafted and functionalized PP were examined by a Perkin Elmer Spectrum One FTIR spectrophotometer in attenuated total reflectance (ATR) mode. The samples were scanned in the range 600–4000 cm^{-1} with a 4 cm^{-1} resolution.

The surface morphology of the pristine, grafted and functionalized PP was investigated by scanning electron microscopy (SEM) using a Hitachi SU3500 (Hitachi, Japan) microscope at acceleration voltage of 5 kV and magnification of 300 $\times$.

Seiko Instruments TG/DTA 6200 ExStar 6000 was used to record the thermal decomposition properties of polymer samples over the temperature range from 25 to 500 $^\circ\text{C}$ with a programmed temperature increment of 10 $^\circ\text{C min}^{-1}$ under dynamic nitrogen atmosphere.

3. Results and discussion

3.1. Grafting and optimization process

A first-order model with interactions mainly estimates the main effects of both the experimental factors and the interaction of the factors with each other. Adding interaction terms and quadratic terms

Table 2

ANOVA test for determining the significance of the variables and goodness of fit of the model.

Source	F-value	Prob > F	Significance
Model	120.63	< 0.0001	Significant
Dose, A	244.39	< 0.0001	Significant
Time, B	625.10	< 0.0001	Significant
Conc, C	88.61	< 0.0001	Significant
AB	85.42	< 0.0001	Significant
B ²	15.39	0.0002	Significant
C ²	19.72	< 0.0001	Significant
R ²	0.9526		
Adjusted R ²	0.9447		
Predicted R ²	0.9340		
Adequate Precision	42.467		

introduces curvature into the response function (Hui, 2006). The quadratic model with interaction and quadratic terms proposed for the response variable, Dg, was based on the quadratic polynomial regression method. Stepwise deletion of terms was applied to eliminate the statistically non-significant terms.

Sixty four (64) combinations of experimental variables were used in the optimization experiments. The independent and dependent variables were fitted to the quadratic model equation and then tested for goodness of fit. The *p-values* were used as a tool to check the significance of each parameter, both individually and interacting. Afterwards, the influence of noise on the data was assessed using ANOVA and the results are shown in Table 2.

The F-values, which is the ratio of noise to response, of both models signified that the models were significant. The values of “Prob > F” or *p-values* less than 0.05 indicate the model terms are significant. All three independent variables i.e. absorbed dose, reaction time and monomer concentration, were significant, implying that Dg is strongly correlated to the reaction parameters. Data from Table 2 also shows that the interaction term between absorbed dose and reaction time (AB), and the quadratic terms for reaction time (B²) and monomer concentration (C²) had significant effects on the response factor, Dg. The response data were fitted to a quadratic polynomial regression equation, in terms of the coded factors with a significance level of $p \leq 0.05$ according to Eq. (4):

$$\text{Dg, \%} = 96.80 + 28.82 A + 46.10 B + 17.36 C + 22.86 AB - 12.13 B^2 - 13.73 C^2 \quad (4)$$

where A = absorbed dose in kGy, B = reaction time in h, C = monomer concentration in % wt/wt.

The goodness of fit of the model was evaluated by the coefficient of determination (R²), adjusted-R², predicted-R² and adequate precision. Table 2 also summarizes the statistics used to test the adequacy of the model. Coefficient of determination is the proportion of variation in the response attributed to the model. A value close to 1 implies a good-fit model (Xiangli et al., 2008). The model for Dg had a satisfactory R² value of more than 94%. However, R² always increases upon the addition of a new variable to the model, regardless of whether the additional variable is significant or not. Therefore, adjusted-R² is more preferred to evaluate the adequacy of the model. The adjusted-R² should be over 90% to indicate high degree of correlation between the observed and predicted values. Based from Table 2, R² and adjusted-R² for the models did not differ largely implying that non-significant terms were excluded from the model. Predicted-R² gives a measure of the amount of variation in the new data based from the model. Generally, a number closer to one is preferred. The adequate precision data measures the signal to noise ratio. Ratios greater than 4 indicate adequate model discrimination.

The normality of the data was checked from the plot of normal percent probability against externally studentized residuals of re-

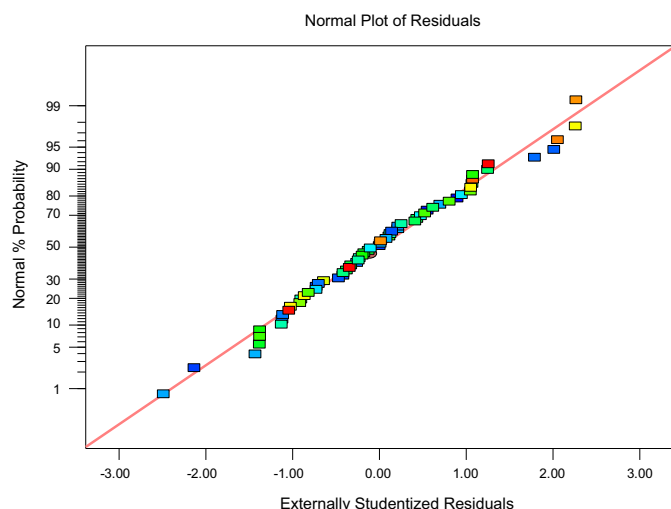


Fig. 1. Normal plot of residuals for experimental runs in the optimization process.

sponses and the result is shown in Fig. 1. A linear correlation was obtained, which means that there was no violation in normality in the data fitting cases.

Fig. 2 shows the main and interaction effects of factors on the response, Dg, at different monomer concentrations. The contour plots indicate a non-linear relationship between reaction time and absorbed dose. The calculated Dg increased with increasing absorbed dose. The interaction of high-energy electrons with PP resulted in the creation of

free radicals, which served as initiation sites for graft polymerization. Higher absorbed dose yielded more free radicals on the surface, which initiated more graft polymer chains leading to higher Dg.

At constant absorbed dose, the Dg increased over time because the monomer molecules had more time to diffuse to the initiation sites and propagating polymer chains. Finally, Dg increased with increasing monomer concentration at a given absorbed dose and reaction time. This is simply because of closer proximity of the GMA molecules with the free radicals on PP and propagating polymer chains.

An optimization study was performed to predict the optimum values of factors for the target adsorbent application. The desirability function was applied using Design Expert 9 software. The goal of the optimization was set to a target Dg of 150% at 5% monomer concentration and both reaction time and absorbed dose, which had values set within the range 1–4 h and 10–40 kGy, respectively, were optimized. The 150% target Dg was found to be sufficient for adsorbent application because after post-grafting reactions, it gave functional group density comparable to commercially available adsorbent resin (Seko et al., 2007). Moreover, grafted PP with very high Dg tends to be brittle and difficult to handle after grafting. The 5% monomer concentration was chosen to minimize the amount of monomer in the emulsion, which would contribute to lower process cost. The 3% monomer concentration was not applicable because the target Dg was not achieved at this concentration even if both absorbed dose and reaction time were maximized, as shown in Fig. 2a.

Fig. 3 shows the Ramp report that summarizes the optimum reaction parameters leading to a predicted Dg of 150%. The calculated optimum parameters were 39.8 kGy absorbed dose and 3.5 h grafting time at 5% monomer concentration. The validity of the calculated

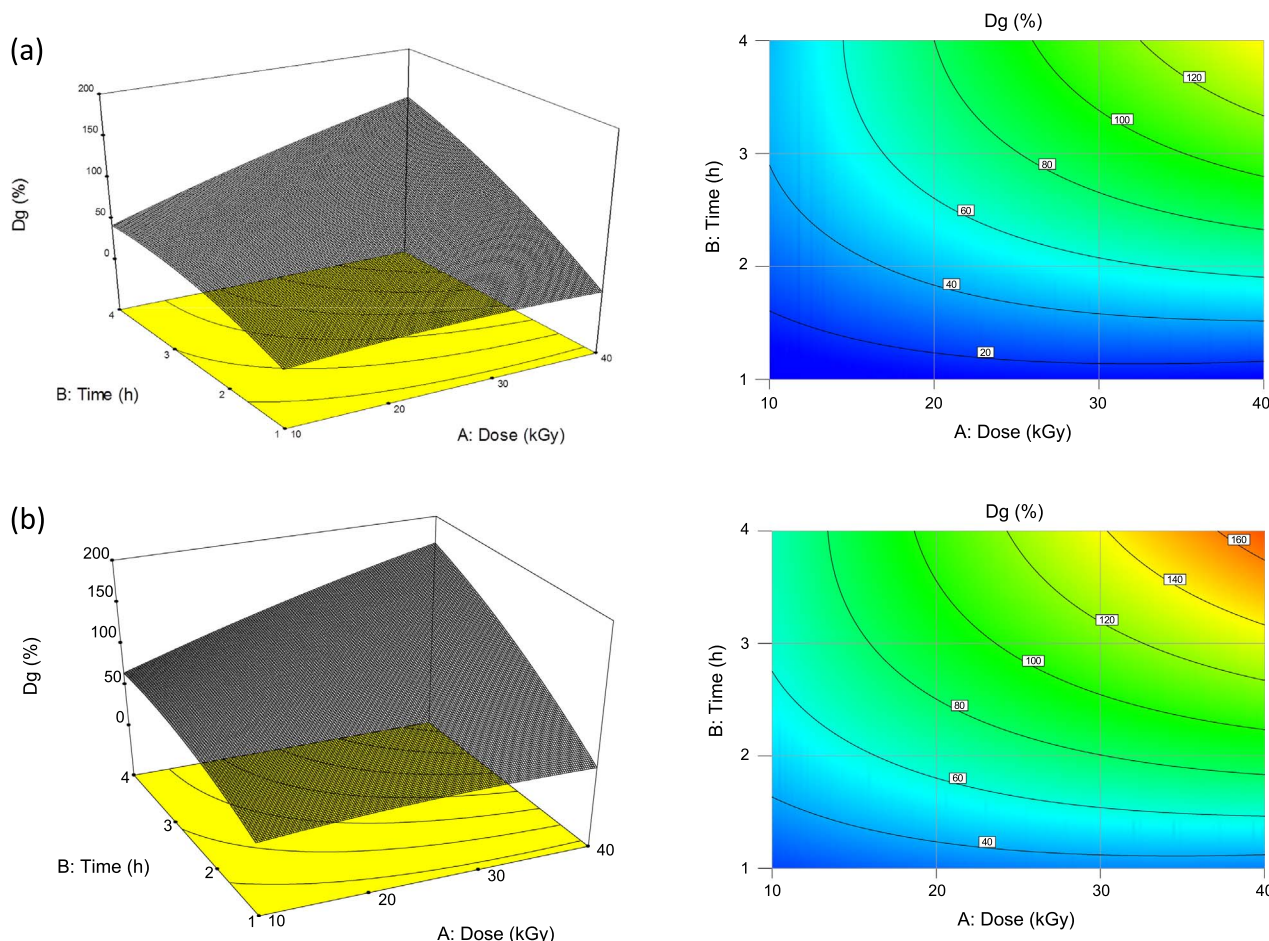


Fig. 2. 3D surface plots and 2D contour plots of Dg as a function of reaction time and absorbed dose at (a) 3%, (b) 5%, (c) 7%, (d) 9% monomer concentration with 5:1 monomer-surfactant weight ratio.

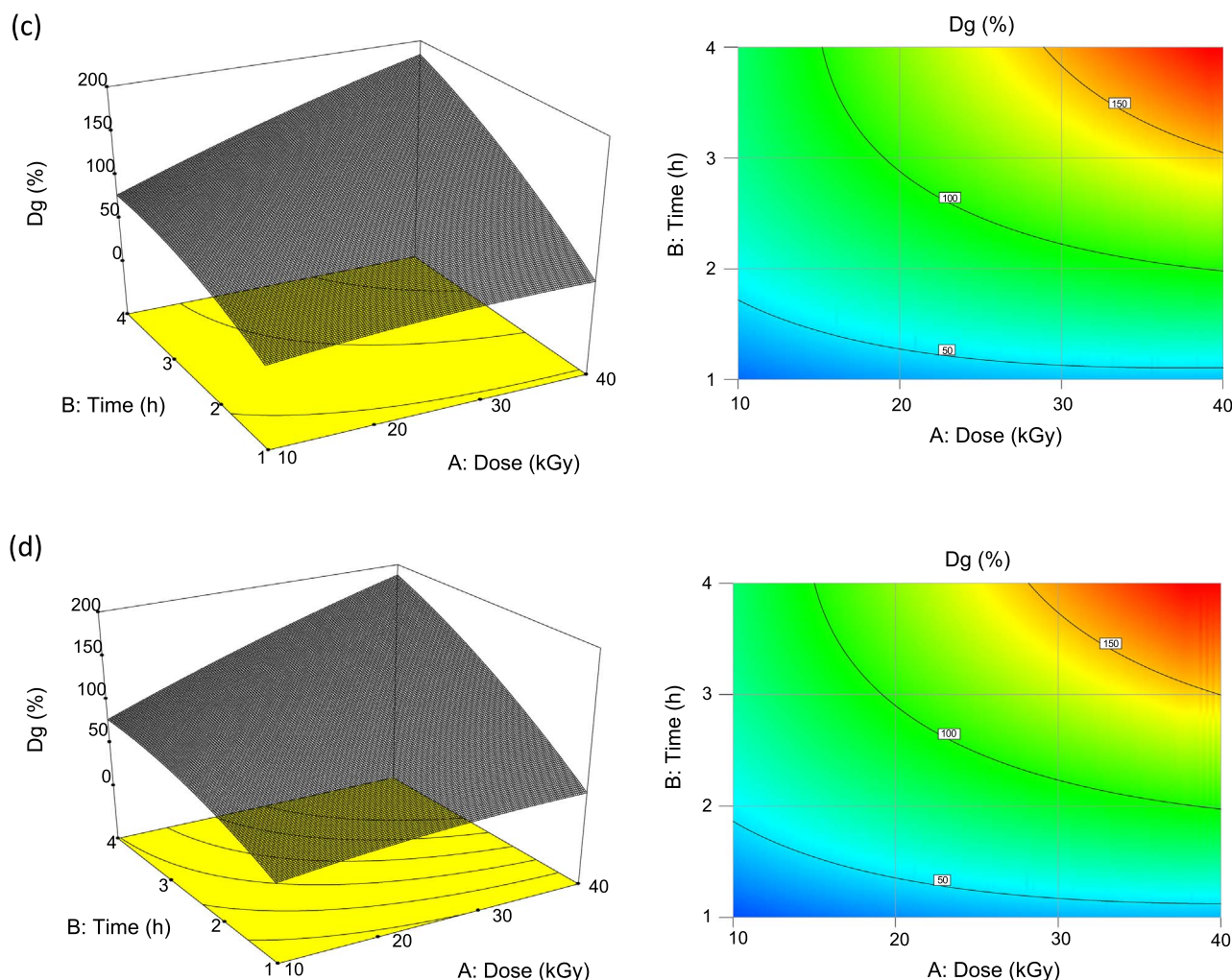


Fig. 2. (continued)

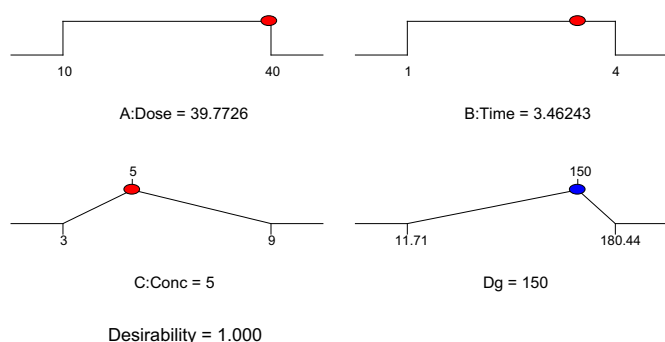


Fig. 3. Ramp report showing the optimum parameters at 1.000 desirability level.

optimum parameters was tested by performing 10 grafting experiments at the optimum grafting conditions. An average experimental Dg value of 153%, with a standard error of 3%, was achieved. The close agreement between the predicted and experimental data provides a strong evidence of the success of the developed model.

3.2. Reaction of PP-g-PGMA with EDA and IDA

The presence of the epoxy group in PGMA allows synthesis of tailor-made grafted materials for various adsorbent applications after post-grafting reactions of PP-g-PGMA with suitable reagents. The PP-g-PGMA samples with approximately 150% Dg were functionalized with

EDA and IDA, resulting to the introduction of amine and carboxylic functional groups by ring opening reaction of the epoxy group from the grafted PGMA. The immobilized functional groups have high affinities towards different metal ions hence the synthesized adsorbents are very suitable for separation applications (Nasef and Guven, 2012). The effect of EDA concentration on the amine group density at different reaction time was studied and the result is shown in Fig. 4. The amine group densities of the functionalized samples increased rapidly and leveled off at 2.5 and 3.0 mmol/gram after 30 min of reaction with 25% and 50% (wt/wt) EDA, respectively. At a given reaction time, the amine

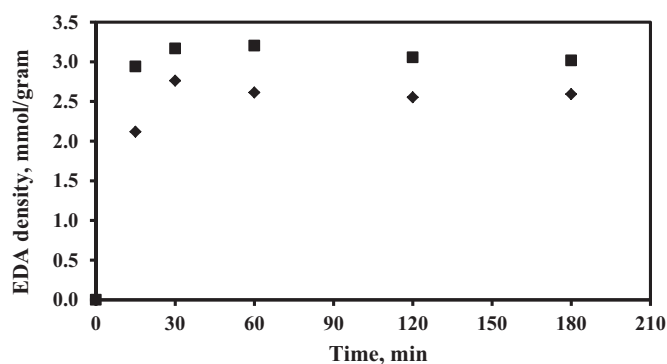


Fig. 4. Amine functional group density of grafted PP after reacting with 25% (♦) and 50% (■) (wt/wt) EDA in isopropanol solvent.

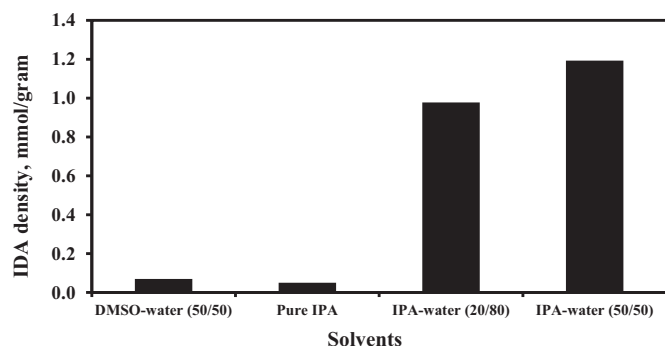


Fig. 5. Calculated iminodiacetic acid functional group density at different solvents.

group density increased with EDA concentration because at higher concentration, more EDA molecules were in close proximity to the epoxy groups from the grafted PGMA, which lead to higher reaction rates. High amine group density material is preferable for separation application because it has more adsorption sites. The optimum conditions for amine functionalization of PP-g-PGMA were therefore determined as 50% (wt/wt) EDA and 30 min reaction time.

Fig. 5 shows the effect of solvent on the calculated IDA functional group density. Two factors affect the choice of solvent for the functionalization reaction. First, the solvent must dissolve IDA at the working temperature and second, the solvent must facilitate transport of the dissolved IDA molecules to the graft chains, hastening the ring-opening reaction of the epoxy group. Although water is a good solvent for IDA, it does solvate the PGMA graft chains because of disparity in polarity; hence, addition of another solvent such as IPA or DMSO was necessary.

Among the solvents studied, IPA-water with 50/50 (wt/wt) composition yielded the highest IDA group density after the 24-h reaction. The data shows that decreasing the amount of IPA resulted to a lower calculated group density probably due to the inefficient solvation of the graft chains. Applying pure isopropanol or aqueous DMSO yielded very low IDA group densities, because of the low solubility of IDA in these solvents. The resulting IDA functional group density of 1.2 mmol/gram was almost similar to the value obtained by Choi and Nho (1999) in their study about the modification of PE hollow fiber membranes with iminodiacetic acid and sulfonic acid groups.

3.3. Effect of initial pH

The amount of metal ions removed by adsorbents from aqueous solutions is dependent on the original pH of the solution (Sekine et al., 2010; Ma et al., 2011; Madrid et al., 2013). To study the effect of pH on the Cr(VI), Cd(II) and Pb(II) uptake by the functionalized PP-g-PGMA, batch adsorption was performed using 50 ppm metal ion solutions with different initial pH. In addition to Cr, which is the main heavy metal ion component of waste waters from leather tanning industry, the target application of the synthesized adsorbents, the researchers also studied the effect of pH on the adsorption of Cd(II) and Pb(II) ions. The pH range studied for batch adsorption experiments was chosen so that it approximated the pH of actual tanning industry wastewater, except for Pb(II) which precipitated at pH higher than 4.

Both functionalized materials exhibited pH dependency during adsorption (Fig. 6). Maximum adsorption of Cr(VI) was achieved using the EDA functionalized PP-g-PGMA at pH 4. At lower pH values, higher H^+ ion concentration resulted to protonation of amine groups on the amine type adsorbent. Additionally, chromium at pH 4.0–6.0 exists in anionic forms such as $HCrO_4^-$ and $Cr_2O_7^{2-}$ (Saha and Orvig, 2010). The strong electrostatic attraction between the protonated

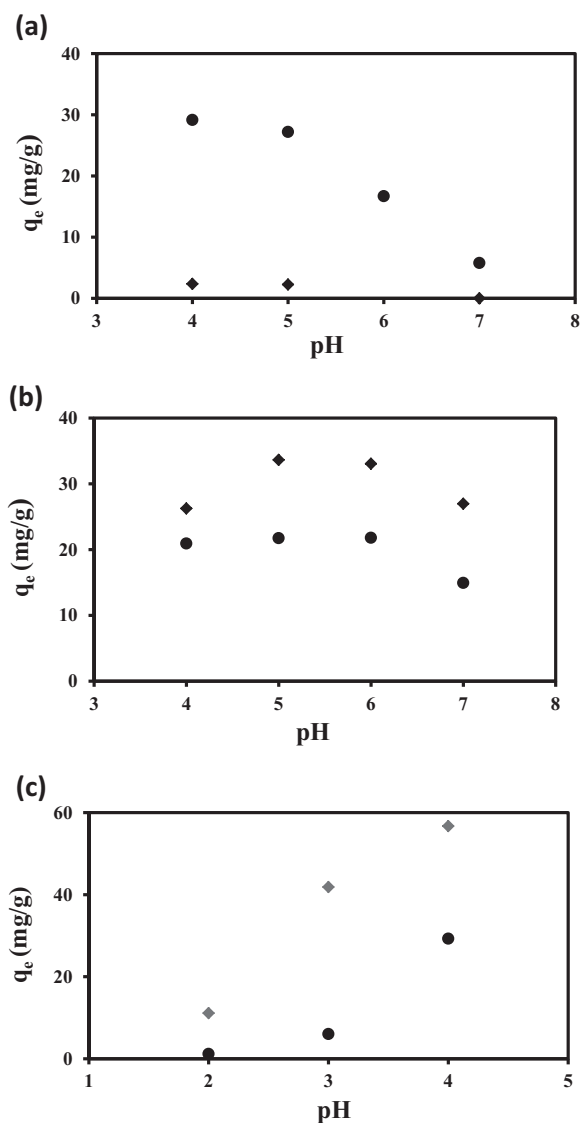


Fig. 6. Effect of pH on the removal of (a) chromium, (b) cadmium, and (c) lead using (♦) IDA and (●) EDA functionalized PP-g-PGMA.

adsorbent surface and the negatively charged chromate ions significantly enhanced the removal efficiency; therefore adsorption was higher at lower pH (Attia et al., 2010). The amount of adsorbed Cr(VI) decreased as the pH of the solution increased. This observation might be due to the competition between OH^- and chromate ions for the protonated amine group sites at higher pH values (Gang et al., 2010; Muthukumaran and Beulah, 2011; Wang et al., 2013). Negligible Cr(VI) sorption by the IDA functionalized material, especially at higher pH, was due to the electrostatic repulsion between the chromate ions and the carboxylate groups of the IDA functionalized PP-g-PGMA.

EDA functionalized PP-g-PGMA showed lesser Cd(II) and Pb(II) sorption than IDA functionalized PP-g-PGMA at the pH range studied. These metals existed in cationic form in the solution; hence the electrostatic repulsion caused by the protonated amine group of the EDA functionalized PP-g-PGMA was more dominant. Maximum sorption by the carboxylic functionalized PP-g-PGMA was achieved at pH 5–6 and 4 for Cd(II) and Pb(II), respectively. The higher extent of ionization of the acidic group of IDA with increasing pH resulted to a more negatively charged surface of the IDA functionalized PP-g-PGMA, which increased the electrostatic attraction between the metal ions and the carboxylic acid functional group of IDA (Atieh et al., 2010; Tahermansouri et al., 2014).

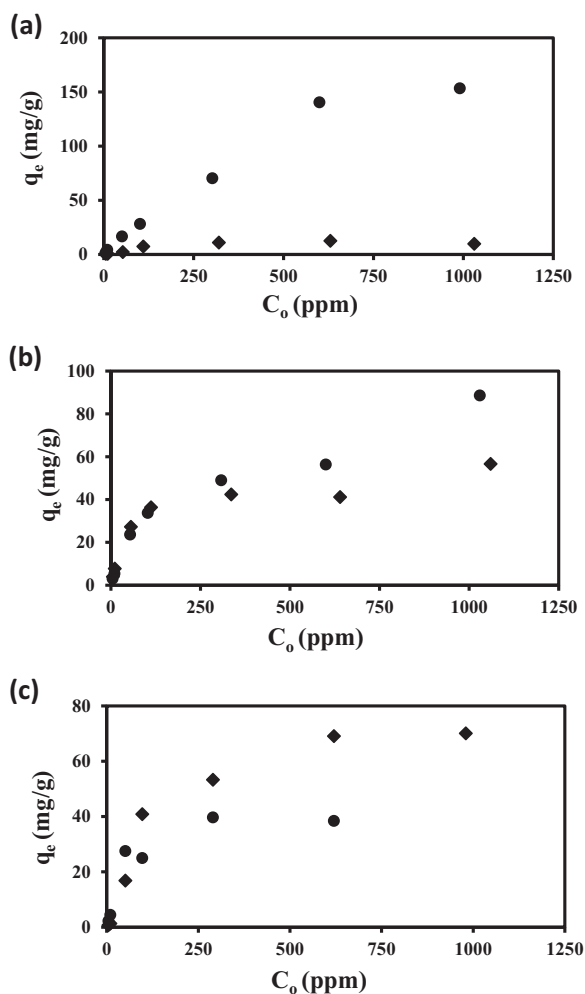


Fig. 7. Effect of initial concentration on (a) chromium, (b) cadmium, and (c) lead adsorption capacities of (♦) IDA and (•) EDA functionalized PP-g-PGMA.

3.4. Effect of initial metal ion concentration

The amount of metal ions adsorbed per unit mass of the EDA and IDA functionalized PP-g-PGMA as a function of initial metal ion concentration is shown in Fig. 7. Generally, the quantity of adsorbate on the surface of the solid phase increased with increasing initial metal ion concentration. The amine type adsorbent showed affinity to all the studied metal ions. Maximum q_e values of 153 mg/g, 89 mg/g and 40 mg/g were calculated for Cr(VI), Cd(II) and Pb(II) uptake by the amine type adsorbent, respectively. However, the amount of removed metal ions relative to the initial amount decreased with increasing initial metal ion concentration. The total number of adsorption sites was limited so increasing the amount of metal ion in the solution resulted to incomplete sorption. This caused the observed decrease in percentage removal when the initial sorbate concentration was increased. A similar trend was shown by the IDA-functionalized PP-g-PGMA, in which maximum q_e values of 13 mg/g, 57 mg/g and 70 mg/g were calculated for removal of Cr(VI), Cd(II) and Pb(II), respectively.

The experimentally evaluated metal ion adsorption capacities of the amine and carboxylic type adsorbents in the present work were comparable to the adsorption capacity values reported in other studies (Wang et al., 2001; Chen et al., 2006; Pehlivan and Altun, 2007).

3.5. Adsorption isotherms

Adsorption isotherm was used in determining the equilibrium relationship distribution of the adsorbate to the adsorbent. Several

models are available in predicting the equilibrium distributions but the most commonly used are the Langmuir and Freundlich isotherms.

Langmuir isotherm assumes a monolayer distribution with finite number of identical sites given by the equation:

$$q_e = \frac{q_{max} b C_e}{1 + b C_e} \quad (5)$$

where C_e (ppm) is the equilibrium concentration of the adsorbate in the solution, q_e (mg/g) is the amount of metal adsorbed per gram of adsorbent; q_{max} (mg/g) is the maximum adsorption at monolayer coverage and b is the Langmuir constant (L/mg).

Freundlich isotherm describes a multilayer distribution in which the amount of the solute adsorbed per amount of adsorbent increases gradually. Given by the equation:

$$q_e = K_f C_e^{1/n} \quad (6)$$

where K_f is the surface adsorption equilibrium constant in (mg/g) and n refers to the interaction between exchange sites in the adsorbent and metal ions. A high value for $n > 1$ indicates favourable adsorption.

The adsorption data for Cr(VI), Cd(II) and Pb(II) using the EDA and IDA functionalized PP-g-PGMA were fitted to the linear forms of Langmuir and Freundlich isotherms. The Langmuir and Freundlich constants and R^2 values for different metal ions are given in Table 3 and Table 4.

The adsorption isotherms for both amine and carboxylic type adsorbents indicate that adsorption of the investigated metal ions increased with increasing equilibrium concentrations. The R^2 data in Table 3 suggest that the removal of Cr(VI) and Cd(II) from the solution can be modelled using the Freundlich equation while the removal of Pb(II) can be approximated by the Langmuir equation. For the IDA functionalized PP-g-PGMA (Table 4), both Cd(II) and Pb(II) adsorption showed excellent fit to the Langmuir equation while Cr(VI) removal can be modelled using the Freundlich equation. Interestingly, the calculated Langmuir constant, q_{max} , for all metals were almost similar to the values obtained from the batch adsorption experiments, validating the results of isotherm fitting. All calculated Freundlich parameter n are greater than 1, indicating favourable adsorption. Data from Tables 3 and 4 show that the amine type adsorbent has high capacity for adsorbing Cr(VI) while the carboxylic type adsorbent has better adsorption capacity for Pb(II). This information corroborated with the data obtained from the experiments on effect of pH on adsorption.

Table 3

Isotherm constants for adsorption of metal ions by EDA functionalized PP-g-PGMA.

Metal ion	LANGMUIR			FREUNDLICH		
	q_{max} (mg/g)	b	R^2	n	K_f	R^2
Chromium	213	0.004	0.93	1.57	2.75	0.99
Cadmium	83	0.008	0.92	3.53	11.48	0.98
Lead	39	0.103	0.99	3.07	6.68	0.84

Table 4

Isotherm constants for adsorption of metal ions by IDA functionalized PP-g-PGMA.

Metal ion	LANGMUIR			FREUNDLICH		
	q_{max} (mg/g)	b	R^2	n	K_f	R^2
Chromium	12	0.008	0.94	1.60	0.20	0.95
Cadmium	53	0.037	0.96	4.96	14.48	0.86
Lead	70	0.049	0.99	7.99	7.88	0.97

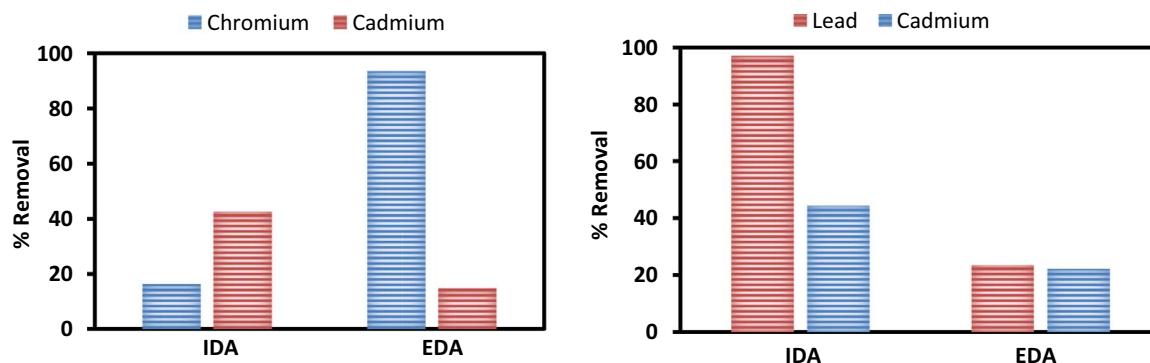


Fig. 8. Competitive adsorption of Cr(VI)/Cd(II) (left) and Cd(II)/Pb(II) (right).

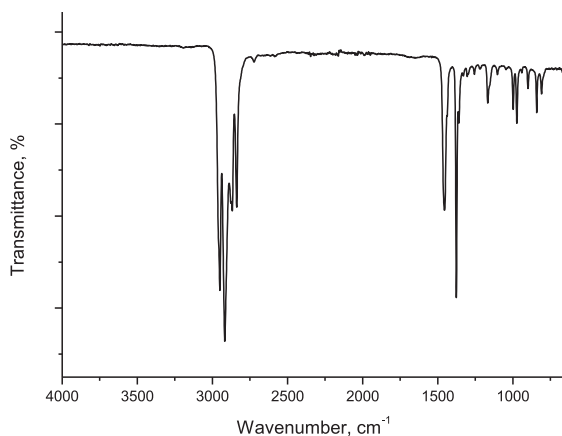
3.6. Competitive adsorption

Binary metal ion systems were applied to determine the selectivity of the synthesized adsorbents. Studying a tertiary metal ion system for competitive adsorption was not possible because of the reaction between chromate and lead ions. The result of competitive adsorption experiments for Cr(VI)/Cd(II) and Pb(II)/Cd(II) pairs is shown in Fig. 8.

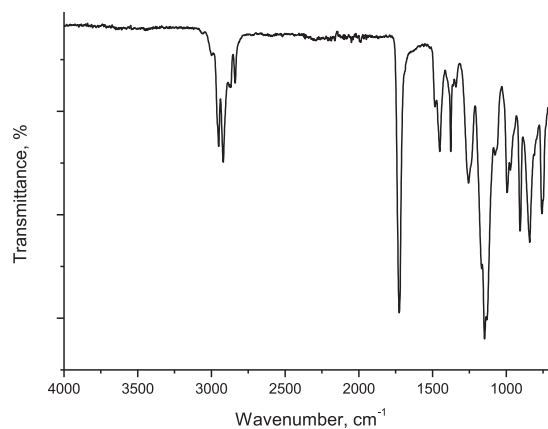
The EDA functionalized PP-g-PGMA showed better affinity for Cr(VI) over Cd(II) ions while it did not exhibit preferential adsorption between Pb(II) and Cd(II) ions. At lower pH, the amine groups were more protonated, therefore the electrostatic attraction between the adsorbent and the metal oxanion form, which in this case Cr(VI) in the form of HCrO_4^- , was enhanced. On the other hand, IDA functionalized PP-g-PGMA showed stronger affinity for Cd(II) over Cr(VI), primarily because of the

repulsion between the carboxylate ion on the adsorbent and the chromate form of Cr(VI). In Pb(II)/Cd(II) system, IDA functionalized PP-g-PGMA exhibited high adsorption for both metal. The preference of the carboxylic type adsorbent for Pb(II) over Cd(II) may be attributed to the higher electronegativity of Pb(II) (2.33) as compared to Cd(II) (1.69) (Balasubramanian et al., 2009). Moreover, Pb(II) is a borderline hard Lewis acid while Cd is a soft Lewis acid, hence Pb(II) has greater affinity for carboxylic acid functional group, which is a hard Lewis base (Appel and Ma, 2002). The binary metal ion experiments showed that the affinities of the synthesized adsorbents towards the studied metal ions at pH 4 were in the following order: Cr(VI) > > Pb(II) ~ Cd(II) for the EDA functionalized PP-g-PGMA; and Pb(II) > Cd(II) > Cr(VI) for the IDA functionalized PP-g-PGMA.

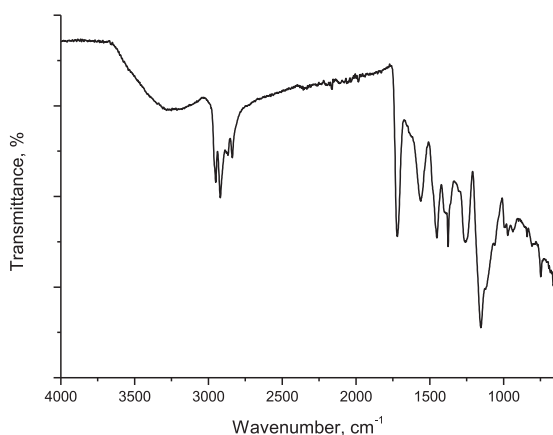
(a) PP nonwoven fabric



(b) PP-g-PGMA, Dg = 180%



(c) EDA functionalized PP-g-PGMA



(d) IDA functionalized PP-g-PGMA

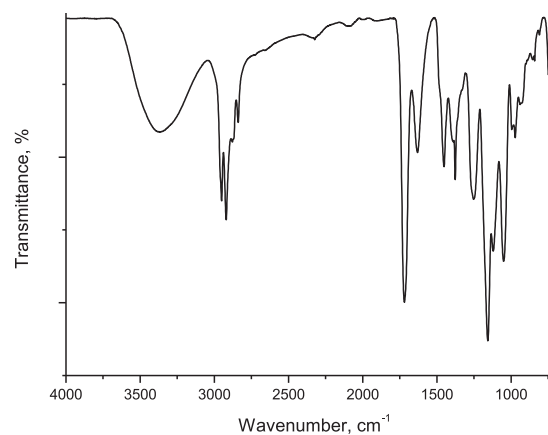


Fig. 9. FTIR-ATR spectra of (a) pristine PP, (b) PP-g-PGMA 180% Dg, (c) PP-g-PGMA reacted with EDA and (d) PP-g-PGMA reacted with IDA.

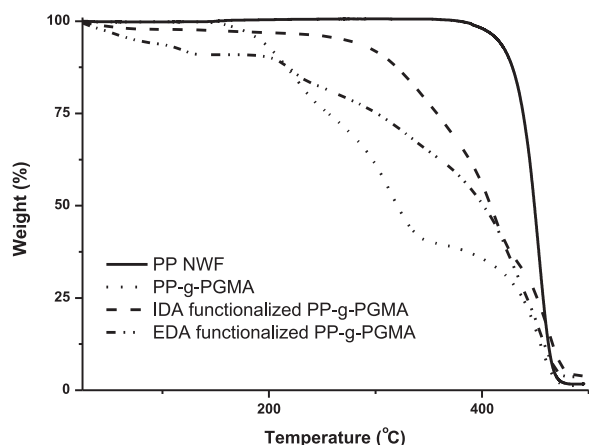


Fig. 10. Thermographs of pristine PP, PP-g-PGMA, EDA and IDA functionalized PP-g-PGMA.

3.7. Characterization of grafted and functionalized materials

3.7.1. FTIR analysis

Graft polymerization of GMA from irradiated PP introduced new chemical bonds which can be detected using FTIR spectroscopy. The ATR-FTIR spectra of the pristine, grafted and functionalized PP are shown in Fig. 9. The FTIR spectrum of the pristine PP trunk polymer has absorption peaks at around $2800\text{--}2900\text{ cm}^{-1}$ and $1400\text{--}1500\text{ cm}^{-1}$ that correspond to C-H stretching and CH_2 bending modes, respectively. After grafting with PGMA, new peaks appeared that correspond to the grafted polymer chains: $\text{C}=\text{O}$ stretch at 1720 cm^{-1} , C-O stretch at 1120 cm^{-1} and the characteristic epoxy group vibrational modes at around 1250 , 900 and 850 cm^{-1} . Besides the characteristic IR absorption frequencies of bonds

from PGMA, peaks from PP were observed from the FTIR spectrum of grafted PP. The peaks that correspond to the epoxy group are almost absent on Fig. 9c and d, which depict the FTIR spectrum of PP-g-PGMA after reacting with EDA and IDA, respectively. After reaction with EDA, new peaks appeared at around 3300 cm^{-1} and 1590 cm^{-1} , corresponding to N-H stretch and N-H bending vibrations. The broad O-H band observed between 3250 and 3500 cm^{-1} at Fig. 9d is due to introduction of carboxylic groups from iminodiacetic acid onto grafted PP. Another sharp peak appeared at around 1640 cm^{-1} which is very characteristic of C-N vibration (Kavakli et al., 2014). The absence of the characteristic peaks of epoxy group and appearance of new peaks characteristic of EDA and IDA confirmed the successful ring-opening reactions and functionalization of PP-g-PGMA.

3.7.2. Thermogravimetric analysis

Graft polymerization of GMA on PP and the subsequent functionalization reactions imparted new chemical bonds that modified the thermal behavior of the PP trunk polymer. Fig. 10 demonstrates the thermal decomposition profile of pristine PP, PP-g-PGMA, EDA and IDA functionalized PP-g-PGMA. The PP base material showed a single decomposition profile with maximum weight loss at around 450°C . In PP-g-PGMA, a three-step decomposition profile was observed, with the last step coinciding with the decomposition profile of the PP trunk polymer. The initial degradation step was attributed to depolymerization from the unsaturated chain ends and the weight loss above 300°C was associated with degradation by random chain scission (Zulfqar et al., 1990). Hence, the observed weight losses at lower temperatures were due to the PGMA graft chains.

The degradation profiles of the amine and carboxylic acid functionalized PP are also shown in Fig. 10. The amine functionalized PP-g-PGMA showed an initial weight loss from 25 to 150°C that can be attributed to loss of adsorbed water from the sample. Afterwards, the material started to degrade from 200 to 500°C . The aminated material

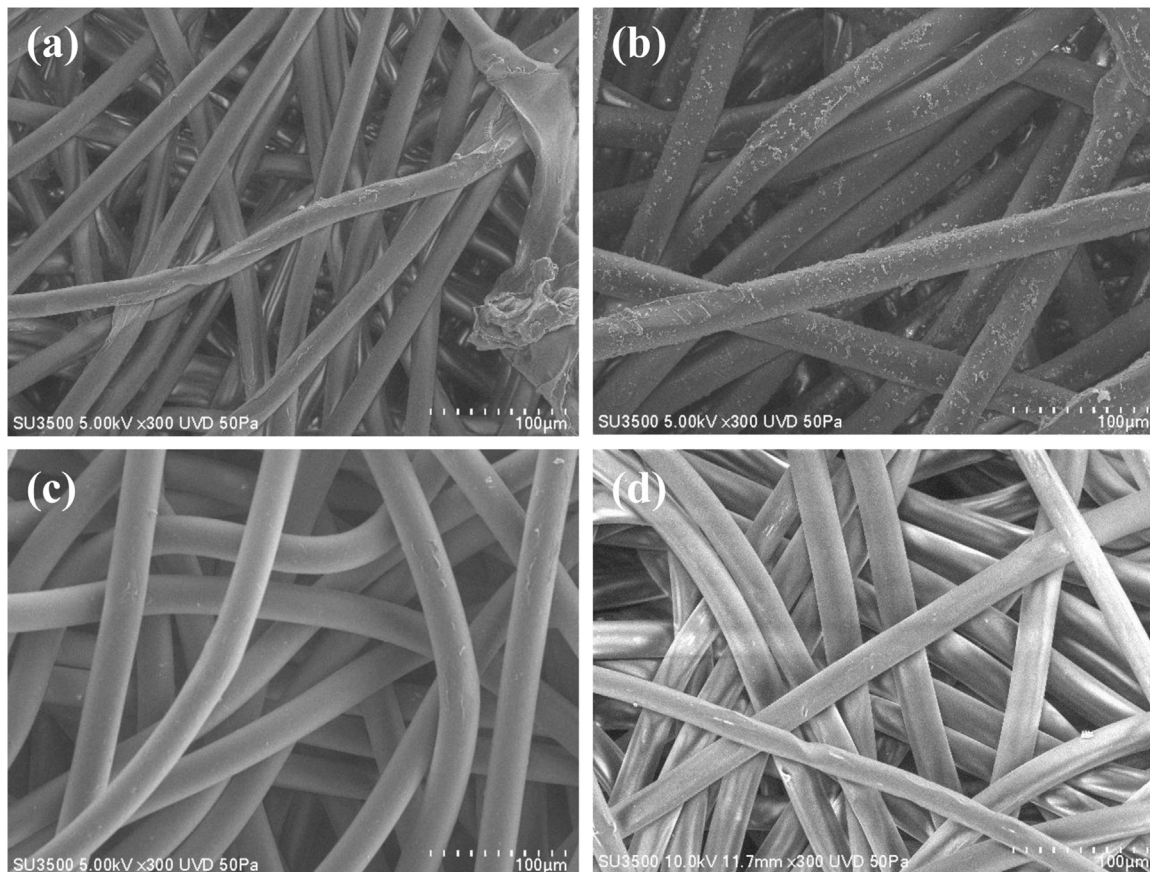


Fig. 11. SEM images of (a) pristine PP, (b) PP-g-PGMA with 180% Dg, (c) PP-g-PGMA reacted with ethylenediamine and (d) PP-g-PGMA reacted with iminodiacetic acid.

showed an improved thermal behavior from 200 to 400 °C compared to PP-g-PGMA, i.e. the decomposition of the amine functionalized PP-g-PGMA occurred at a slower rate over the indicated temperature range. This observation can be attributed to the presence of terminal $-NH_2$ that were imparted to the PP after the functionalization step with ethylenediamine. During thermal decomposition of the aminated material, $-NH_2$ acted as a radical scavenger (Zaharescu et al., 2000) that slowed down the degradation of amine functionalized PP-g-PGMA in nitrogen atmosphere. The thermograph of IDA functionalized PP-g-PGMA showed minor weight loss at the start of the analysis. This weight loss can also be due to the evolution of physically adsorbed water. This was followed by multi-step degradation which was due to random degradation of the polymer chain. Moreover, IDA-functionalized PP-g-PGMA exhibited lower weight loss compared to PP-g-PGMA on the temperature range 200–400 °C. Hence, the thermal stability of the grafted materials improved after the reaction with IDA.

3.7.3. SEM analysis

Fig. 11 shows the SEM images of the pristine, grafted and amine functionalized PP. Pristine PP is porous and consisted of fibers with homogeneous morphology, except for some damages that were imparted during the manufacturing process of the fibers. Significant changes in the fibers' dimension and morphology were observed after grafting. Grafting with PGMA increased the fiber diameter and resulted to a rougher surface morphology. However, it must be noted that although the surface of the fibers became slightly rougher after the grafting process, no homopolymers were observed on the surface of the grafted material based from the SEM pictures. Both EDA and IDA functionalized PP-g-PGMA had almost similar fiber size to PP-g-PGMA but the functionalized materials had smoother fiber surfaces than grafted PP. This indicates that the reaction of EDA and IDA with PP-g-PGMA did not result in any damage to the material, making it a suitable method for introducing amine and iminodiacetic acid groups to the olefinic trunk polymer.

4. Conclusions

The electron beam-induced pre-irradiation graft polymerization of GMA from PP *in vacuo* was successfully conducted under controlled reaction parameters, which were optimized and studied using the full factorial design of experiment. Using ANOVA, the linear terms of absorbed dose, reaction time and monomer concentration, quadratic terms of time and concentration, and interaction term between absorbed dose and time were established as significant independent parameters in the mathematical model that relates Dg to the reaction parameters. The optimum reaction parameters for achieving 150% Dg PP-g-PGMA are 39.8 kGy absorbed dose and 3.5 h grafting time at 5% monomer concentration, 5:1 monomer to surfactant ratio and 40 °C reaction temperature. The FTIR spectra, thermographs and SEM pictures of pristine, grafted and functionalized PP indicated that the modification of the PP surface by grafting of PGMA and functionalization of the epoxy groups were successfully conducted. The batch adsorption data were fitted into Langmuir and Freundlich isotherms. The ion-exchange process showed maximum removal of Cr(VI) and Pb(II) at pH 4 by the amine type and carboxylic type adsorbents, respectively, and Cd(II) in the pH range 5–6 with the carboxylic type adsorbent. The affinities of the synthesized adsorbents towards the studied metal ions were found to be in the following order: Cr(VI) > Pb(II) ~ Cd(II) for the EDA functionalized PP-g-PGMA; and Pb(II) > Cd(II) > Cr(VI) for the IDA functionalized PP-g-PGMA.

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Research and Development, Department of Science and Technology.

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